

5. Features of Sparse Connections:

sparse connections reduce the total num of weights, thereby, reducing the num of weights to store, the num of weights to adjust, and the overall num of computation.

6. Features of Weight Sharing:

- A. weight sharing reduce the num of unique weights, thereby, reducing the num of weights to store, the num of weights to adjust, however, it doesn't reduce the num of computation, because different input values are fed into each neuron in a convolutional layer.
- B. although the storage of weights themselves is reduced due to weight sharing, we still need to storage the activation function values of each neuron in a convolutional layer.

7. Core Advantage of A Convolutional Layer:

a convolutional layer effectively combines the benefits of both sparse connections and weight sharing.

8. Transition from Shallow to Deep Layers:

A. transition flow: shallow convolutional layers → deep convolutional layers → terminal fully connected layer(s).

B.

Layer type	Spatial resolution ($W \times H$)	Number of channels / depth	Forward computation (MACs/FLOPs) status	Number of unique weights
Early convolutional layers (such as input layer)	extremely high (due to input image size)	very small (typically 1 or 3)	heavy load (per fixed operation counting)	very small (due to weight sharing)
Mid-to-late convolutional Layers	progressively decreased	significant increase (into the hundreds)	significant load	moderate load
Final fully connected Layers	no spatial dimension (1d vectors)	channel-to-vector mapping (e.g., flattened)	very small (single matrix-vector multiply)	extremely high (no weight sharing)

9. The Rationale for Using Fully Connected Layers at The End of A CNN:

the core task of the layer(s) at the end of a CNN is to make a definitive classification based on global feature in the image, therefore, the fully connected layer(s) ability to access information from all regions in the image play a critical role at the end of a CNN.